Linear Transformations — 80/20 Interview Notes

1. What is a Linear Transformation? ⭐ ⭐ ⭐

Definition

A function

$$f: U \to V$$

is a **linear transformation** if, for any vectors $x, y$and scalars a, b,

$$\boxed{f(ax + by) = af(x) + bf(y)}$$

This means the transformation preserves:

1. Vector addition
2. Scalar multiplication

The PDF explicitly presents this as the defining property of linear transformations.

Interview intuition

Think:

A linear transformation changes vectors while preserving their linear relationships.

For example:

Original vector space

↓

[Linear

Transformation]

↓

Transformed vector space

A neural-network layer can be viewed as transforming data into a new feature space; the PDF specifically connects linear transformations with ML layers.

Most important shortcut

For a function to be linear:

$$\boxed{f(0) = 0}$$

must be true.

If:

$$f(0) \neq 0$$

then it is **definitely not linear**.

Example — Linear

$$f(x) = 3x$$

Check:

$$f(ax + by) = 3(ax + by)$$
$$= 3ax + 3by$$
$$= af(x) + bf(y)$$

Therefore linear.

Example — Not linear

$$f(x) = x + 5$$

because

$$f(0) = 5 \neq 0$$

So it isn't linear.

The PDF uses translation $f(x) = x + v$, where $v \neq 0$, as an example of a non-linear transformation for exactly this reason.

Interview question

Q: How can you quickly determine whether a transformation is linear?

Answer:

I check whether it preserves addition and scalar multiplication. A quick necessary test is $f(0) = 0$. If $f(0) \neq 0$, it cannot be linear.

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2. Linear Transformation ↔ Matrix ⭐ ⭐ ⭐

This is probably the **most important concept in the chapter**.

Every linear transformation can be represented using a matrix once bases are chosen.

The familiar form is:

$$\boxed{f(x) = Ax}$$

Diagram

A

x ——————→ Ax

| |

| |

Input Output

vector vector

For example:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

and

$$x = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Then

$$Ax = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 10 \\ 11 \end{bmatrix}$$

So:

[3,4] —A—→ [10,11]

Critical interview point

The columns of a matrix tell us where the basis vectors go.

If

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

then

$$Ae_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

and

$$Ae_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Therefore:

Column 1 → image of e_1

Column 2 → image of e_2

The PDF emphasizes this relationship directly.

Interview question

Q: What does each column of a matrix represent?

Each column represents the image of the corresponding basis vector under the linear transformation.

That is a **very good interview answer**.

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3. Why Matrix Multiplication Works ⭐⭐⭐

This is much more important than memorizing the multiplication formula.

Suppose:

$$x \xrightarrow{B} Bx \xrightarrow{A} A(Bx)$$

Therefore:

$$\boxed{ABx = A(Bx)}$$

So:

$$\boxed{AB}$$

represents **composition of transformations**.

The PDF explains matrix multiplication precisely from this perspective: first apply B, then apply A.

Diagram

x

|

▼

[B]

|

▼

Bx

|

▼

[A]

|

▼

ABx

Therefore:

$$\boxed{AB = A \circ B}$$

Important consequence

Matrix multiplication is generally **not commutative**:

$$AB \neq BA$$

because:

B then A

is usually different from:

A then B

Dimension rule

If:

$$A \in R^{n \times m}$$

and

$$B \in R^{m \times l}$$

then:

$$AB \in R^{n \times l}$$

The number of columns of Amust match the number of rows of B.

Interview question

Q: Why is matrix multiplication defined the way it is?

Strong answer:

Because matrix multiplication represents composition of linear transformations. Applying $B$and then Agives $A(Bx) = (AB)x$. The multiplication dimensions therefore correspond to whether the two transformations can be composed.

---

4. Identity Matrix ⭐ ⭐

The identity transformation does nothing:

$$f(x) = x$$

Its matrix is:

$$\boxed{I}$$

For example:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

and:

$$Ix = x$$

Important properties:

$$AI = IA = A$$

The PDF identifies the identity transformation as one of the fundamental matrix representations.

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5. Invertibility and Inverse Matrix ⭐ ⭐ ⭐

Definition

A transformation f is invertible if another transformation f^{-1} exists such that:

$$f^{-1}\big(f(x)\big) = x$$

and

$$f\big(f^{-1}(y)\big) = y$$

In other words:

x —f—→ y

↑ |

└—f⁻¹———┘

The PDF defines invertibility in exactly this way.

For a matrix:

$$\boxed{A^{-1}A = AA^{-1} = I}$$

Why is it useful?

Consider:

$$Ax = b$$

If A^{-1} exists:

$$A^{-1}Ax = A^{-1}b$$

Therefore:

$$\boxed{x = A^{-1}b}$$

This connection to solving systems of linear equations is explicitly introduced in the chapter.

When is a matrix invertible?

For a square matrix:

$$\boxed{\det(A) \neq 0}$$

The PDF states this equivalence explicitly.

Also:

$$\boxed{\ker(A) = \{0\}}$$

for injectivity.

And for a square matrix:

invertible

⇕

injective + surjective

⇕

ker(A) = {0} AND im(A) = V

⇕

det(A) ≠ 0

This is one of the **highest-value relationships to memorize**.

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6. Kernel ⭐ ⭐ ⭐

Definition

The **kernel** of f is:

$$\boxed{\ker(f) = \{x : f(x) = 0\}}$$

For a matrix:

$$\boxed{\ker(A) = \{x : Ax = 0\}}$$

The PDF defines kernel and image together and shows their connection with invertibility.

Intuition

Kernel answers:

Which input vectors get destroyed/mapped to zero?

Diagram:

Input space

|

| A

▼

Output space

x_1 —A—→ 0

x_2 —A—→ 0

x_3 —A—→ non-zero

All inputs that end up at zero belong to the kernel.

Key theorem

$$\boxed{A \text{ injective } \iff \ker(A) = \{0\}}$$

The PDF states this explicitly.

Interview question

Q: What does the kernel tell us?

It tells us which input vectors are mapped to the zero vector. If the kernel contains anything other than zero, the transformation is not injective.

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7. Image ⭐ ⭐ ⭐

Definition

The image is:

$$\boxed{\text{im}(f) = \{f(x) : x \in U\}}$$

For a matrix:

$$\boxed{\text{im}(A) = \{Ax : x \in U\}}$$

The PDF defines image as the set of all vectors that can actually be produced by the transformation.

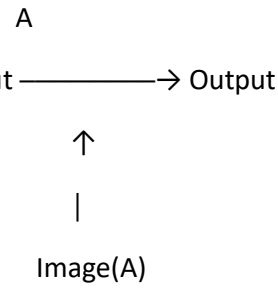
Intuition

Kernel:

What inputs disappear?

Image:

What outputs can actually be produced?



Key theorem

$$\boxed{A \text{ surjective} \iff \text{im}(A) = V}$$

Therefore:

$$\boxed{A \text{ bijective} \iff \ker(A) = \{0\} \text{ and } \text{im}(A) = V}$$

The chapter explicitly states these relationships.

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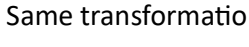
8. Change of Basis ⭐⭐⭐

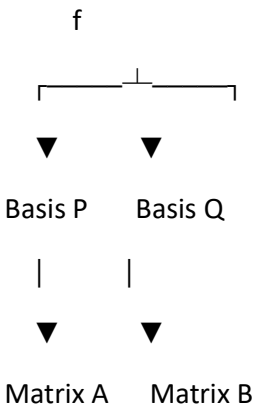
This is a more advanced interview concept, but important for ML/linear algebra.

The **same linear transformation can have different matrix representations depending on the basis**.

The PDF emphasizes that matrix representation depends on the chosen basis.

Conceptual diagram





The transformation hasn't changed.

Only its **representation** changed.

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9. Similar Matrices ⭐⭐

Two matrices $A$and $B$are similar if:

$$\boxed{B = T^{-1}AT}$$

where $T$is an invertible change-of-basis matrix.

The PDF defines similarity this way.

Why should you care?

Because similar matrices represent the **same linear transformation under different bases**.

And a suitable basis can sometimes turn a complicated matrix into a much simpler diagonal matrix. The PDF demonstrates this with:

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

becoming

$$\begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

under a suitable basis.

This leads directly toward **eigenvalues, eigenvectors, and diagonalization**.

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10. Geometric Linear Transformations ⭐ ⭐ ⭐

The chapter focuses on five useful transformations:

1. Stretching
2. Rotation
3. Shearing
4. Reflection
5. Orthogonal projection

These are especially relevant to computer vision and ML; the PDF specifically mentions image augmentation applications.

10.1 Stretching

Matrix:

$$\boxed{A = \begin{bmatrix} c_1 & 0 \\ 0 & c_2 \end{bmatrix}}$$

Example:

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

Then:

$$(x, y) \rightarrow (2, x3y)$$

So:

x direction → ×2

y direction → ×3

The PDF gives this matrix form explicitly.

---

11. Rotation ⭐ ⭐ ⭐

The 2D rotation matrix is:

$$\boxed{R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}}$$

The PDF derives this by looking at where the basis vectors move.

Example: 90° rotation

For:

$$\theta = 90^\circ$$

we get:

$$R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Therefore:

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Important interview property

$$\boxed{R_\alpha R_\beta = R_{\alpha+\beta}}$$

This is actually one of the exercises in the PDF.

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12. Shearing ⭐ ⭐

X-direction shear:

$$\boxed{S_x = \begin{bmatrix} 1 & k_x \\ 0 & 1 \end{bmatrix}}$$

Y-direction shear:

$$\boxed{S_y = \begin{bmatrix} 1 & 0 \\ k_y & 1 \end{bmatrix}}$$

The PDF provides these forms.

Intuition:

Rectangle

↓

Shear

↓

Parallelogram

It changes the shape without simply scaling both axes.

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13. Reflection ⭐ ⭐

Reflection over the x-axis:

$$\boxed{R_x = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}}$$

Reflection over the y-axis:

$$\boxed{R_y = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}}$$

The chapter discusses reflection as a transformation that changes orientation.

Example:

$$(x, y) \to (x, -y)$$

is reflection across the x-axis.

---

14. Orthogonal Projection ⭐ ⭐ ⭐

Projection is especially important conceptually.

Projection of $x$onto y:

$$\boxed{\text{proj}_y(x) = \frac{\langle x, y \rangle}{\langle y, y \rangle} y}$$

The PDF derives the matrix representation:

$$\boxed{P_y = \frac{1}{\| y \|^2} y y^T}$$

so:

$$\boxed{P_y x = \frac{1}{\| y \|^2} y y^T x}$$

Intuition

x

↘

↘

↘

————●————→ y

projection

You're taking the component of $x$that lies along y.

Important insight

Projection can reduce dimension.

For example:

R^2

|

| projection

▼

Line span(y)

The PDF points out that the image of the plane under projection can become a one-dimensional subspace.

This is highly relevant to dimensionality reduction.

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15. Determinant ⭐ ⭐ ⭐

This is another **must-know interview concept**.

For a 2×2matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

the determinant is:

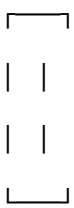
$$\boxed{\det(A) = ad - bc}$$

The PDF interprets determinant geometrically as signed area scaling.

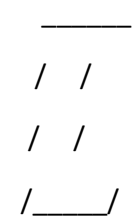
Geometric intuition

Suppose:

Unit square



Apply A:



The square becomes a parallelogram.

The determinant tells us how much the area changes.

$$\boxed{\text{new area} = |\det(A)| \times \text{old area}}$$

The PDF gives exactly this area-scaling relationship.

---

16. What Does the Sign of Determinant Mean?

Positive

$$\det(A) > 0$$

Orientation is preserved.

Negative

$$\det(A) < 0$$

Orientation is reversed.

Zero

$$\det(A) = 0$$

The transformation collapses dimension.

For example:

2D plane

|

| A

▼

1D line

The PDF explains that zero determinant corresponds to zero volume and therefore linearly dependent columns.

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17. Determinant and Invertibility ⭐ ⭐ ⭐

This is one of the **most important formulas to remember**:

$$\boxed{A \text{ invertible } \iff \det(A) \neq 0}$$

If:

$$\det(A) = 0$$

then $A$is singular and has no classical inverse.

The chapter states this theorem explicitly.

Interview explanation

A zero determinant means the transformation collapses volume into a lower-dimensional space. Information is lost, so different inputs can map to the same output. Therefore the transformation cannot be inverted.

That's a much stronger answer than simply saying:

"Because determinant is zero."

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18. Determinant and Linear Independence ⭐ ⭐ ⭐

For a square matrix:

$$\boxed{\det(A) \neq 0 \iff \text{columns of } A \text{ are linearly independent}}$$

Therefore:

det(A) ≠ 0

↓

columns independent

↓

ker(A) = {0}

↓

injective

↓

invertible (square matrix)

And:

det(A) = 0

↓

columns dependent

↓

non-zero vector x exists with Ax = 0

↓

not injective

↓

not invertible

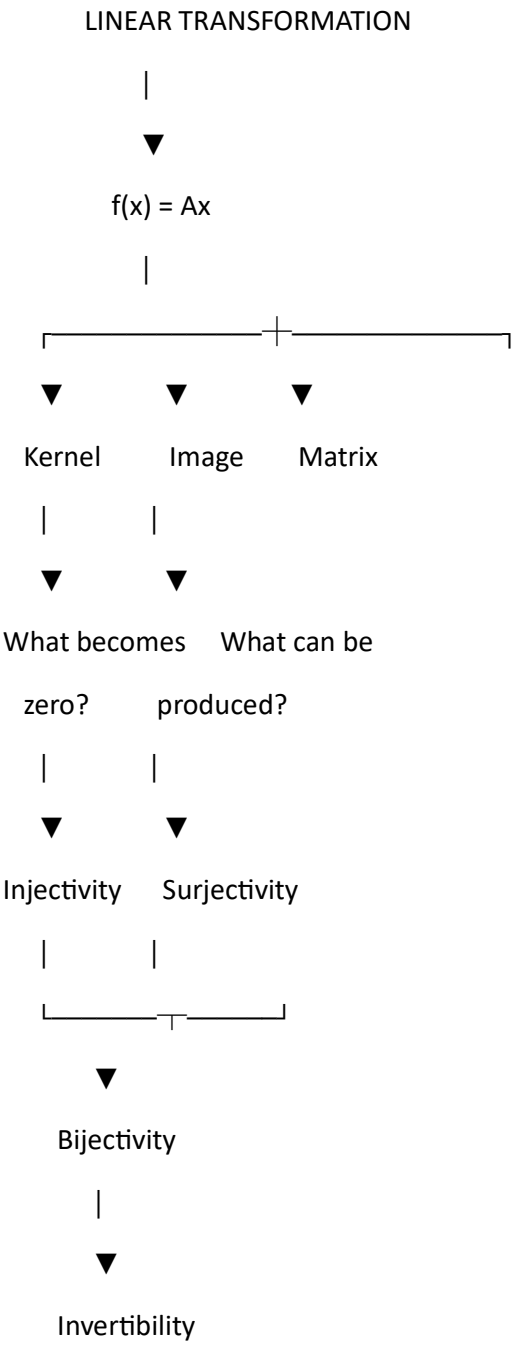
This is the conceptual chain I would memorize for an interview.

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19. The Most Important Mental Model ⭐ ⭐ ⭐ ⭐ ⭐

Don't memorize these as disconnected formulas.

Memorize this:



|

▼

det(A) ≠ 0

This single diagram covers a **large portion of the chapter**.

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20. ML Connection ⭐ ⭐ ⭐

The chapter's most important ML insight is:

A matrix isn't just a table of numbers—it represents a transformation of the feature space.

A neural network layer can be viewed as transforming data:

Input features

|

▼

Linear transformation

|

▼

New feature representation

|

▼

Another transformation

|

▼

Higher-level representation

The PDF explicitly describes neural networks as functions composed of layers that transform data into new feature spaces and identifies linear transformations as key components.

Geometrically, a layer can:

Stretch

+

Rotate

+

Shear

+

Reflect

+

Project

The chapter explicitly connects these transformations to ML and computer vision.

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21. Implementation — Only What Is Worth Implementing

For an ML/AI interview, you **do not need to implement an entire linear algebra library**.

But you should understand these operations.

Matrix-vector multiplication

```
def mat_vec(A, x):
    return [
        sum(A[i][j] * x[j] for j in range(len(x)))
        for i in range(len(A))
    ]
```

Example:

```
A = [
    [2, 1],
    [1, 2]
]


x = [3, 4]


print(mat_vec(A, x))
```

Result:

[10, 11]

Conceptually:

$$Ax = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 10 \\ 11 \end{bmatrix}$$

Matrix multiplication

```
def mat_mul(A, B):
    rows = len(A)
    cols = len(B[0])
    inner = len(B)


    return [
        [
            sum(A[i][k] * B[k][j] for k in range(inner))
            for j in range(cols)
        ]
        for i in range(rows)
    ]
```

The important thing isn't memorizing Python syntax.

Understand:

$$\boxed{ABx = A(Bx)}$$

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22. Interview Examples You Should Be Able to Solve

Example 1 — Is this linear?

$$f(x, y) = (2, x + y \; x - 3y)$$

Yes.

It can be represented as:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & -3 \end{bmatrix}$$

and:

$$f(x) = Ax$$

Example 2 — Is this linear?

$$f(x, y) = (2, x + y + 5\ x - 3y)$$

No.

Because:

$$f(0, 0) = (5, 0)$$

instead of:

$$(0, 0)$$

Example 3 — Calculate determinant

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$
$$\det(A) = 2(2) - 1(1)$$
$$= 3$$

Therefore:

$$\boxed{\det(A) = 3 \neq 0}$$

So $A$is invertible.

This same matrix is used in the PDF's change-of-basis example.

Example 4 — Is this invertible?

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$
$$\det(A) = 1(4) - 2(2) = 0$$

Therefore:

$$\boxed{\text{Not invertible}}$$

Why?

Because the columns are linearly dependent:

$$\begin{bmatrix}2\\4\end{bmatrix} = 2\begin{bmatrix}1\\2\end{bmatrix}$$

Information has been lost.

---

23. 80/20 Priority for Your Interview

If your time is limited, study in this order:

Priority	Concept	Importance
P0	Linear transformation intuition	⭐⭐⭐⭐⭐
P0	Matrix as $f(x) = Ax$	⭐⭐⭐⭐⭐
P0	Matrix columns = transformed basis vectors	⭐⭐⭐⭐⭐
P0	Matrix multiplication = composition	⭐⭐⭐⭐⭐
P0	Kernel and image	⭐⭐⭐⭐⭐
P0	Invertibility	⭐⭐⭐⭐⭐
P0	Determinant ↔ invertibility	⭐⭐⭐⭐⭐
P1	Geometric interpretation of determinant	⭐⭐⭐⭐
P1	Rotation/stretching/projection	⭐⭐⭐⭐
P1	Change of basis	⭐⭐⭐⭐
P1	Similar matrices	⭐⭐⭐
P2	Full determinant derivation	⭐⭐
P2	Formal proofs	⭐
P2	Multilinearity proof	⭐

The PDF itself places substantial emphasis on the conceptual relationship between transformations, matrices, invertibility, basis, and determinants, so this prioritization keeps the high-value material while avoiding unnecessary proof memorization.

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24. 10 Interview Questions You Should Master

Before your interview, make sure you can answer these **without looking at notes**:

1. **What is a linear transformation?**
2. **How do you determine whether a function is linear?**
3. **Why can every linear transformation be represented by a matrix?**
4. **What does each column of a matrix represent?**
5. **Why does matrix multiplication correspond to composition?**
6. **What is the kernel of a transformation?**
7. **What is the image of a transformation?**
8. **What is the relationship between kernel and injectivity?**
9. **What does the determinant mean geometrically?**
10. **Why does** $\det(A) \neq 0$**imply that** A**is invertible?**

The one chain to remember

$$\det(A) \neq 0 \iff \text{columns independent} \iff \ker(A) = \{0\} \iff A \text{ injective} \iff A \text{ invertible}$$

for a square matrix, with the corresponding surjectivity/bijectivity equivalences.

Final Interview Cheat Sheet

LINEAR TRANSFORMATION

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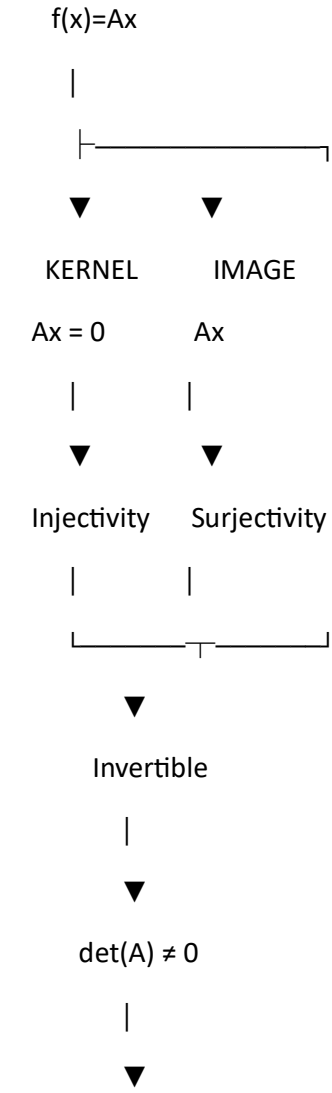
▼

f(ax + by)

= af(x) + bf(y)

|

▼



And geometrically:

Matrix A

- Stretch
- Rotate
- Shear
- Reflect
- Project

▼

Transform feature space

That is the **80/20 core of this chapter**: understand transformations geometrically, connect them to matrices algebraically, and then connect **kernel/image/determinant/invertibility** into one mental model. The chapter's own summary makes this same conceptual jump from matrices to transformations and highlights their relevance to neural-network feature spaces.